

SOUPS 2026 · HANOVER · LIGHTNING TALKS

The end of human heuristics in phishing defence?

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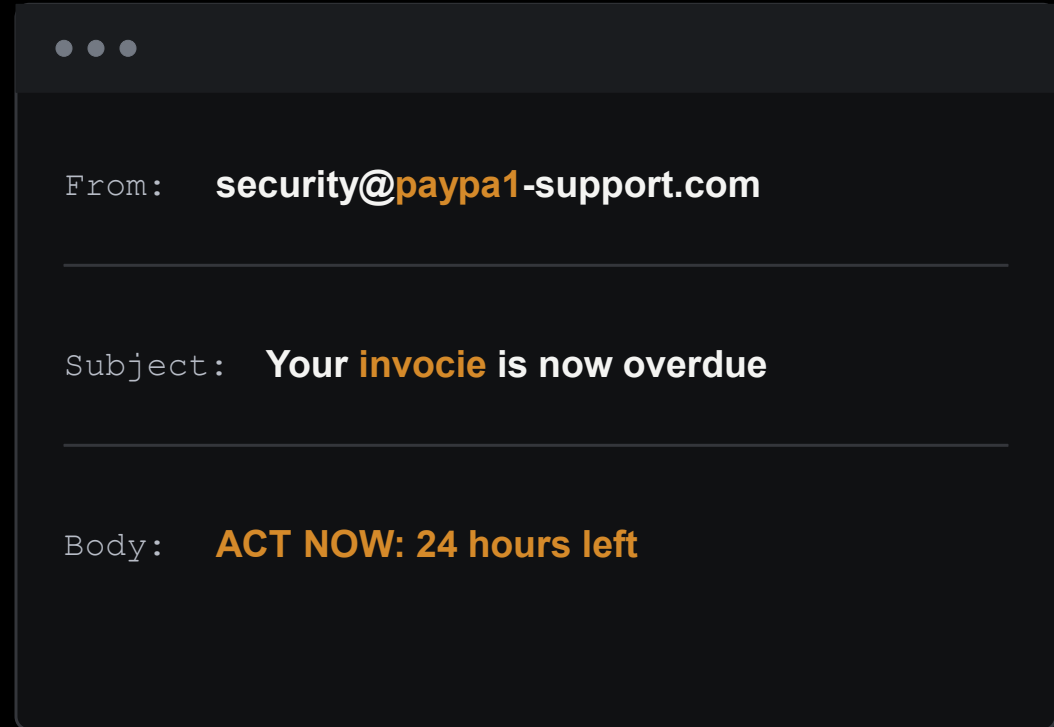
Phishing Training

☐ Check the URL

☐ Look for typos

☐ Beware of urgency

Reasonable advice, but classifiers (can) easily catch these now.



Spam Filters

“This message might be phishing”

“This message is external”

GENERATIVE AI

scalable and sophisticated

DEFENSIVE AI

catching the obvious

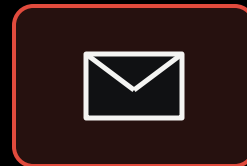
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WHAT REACHES THE INBOX?



ambiguous cases

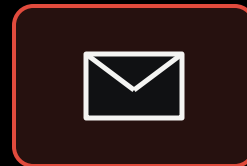
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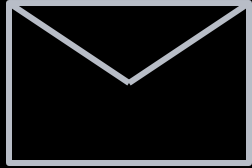
Who catches what gets through?



Filters (can) catch the obvious

Humans deal with ambiguity alone

TODAY



Ask users to classify

Hide reasoning in a black box



NEW



Let AI investigate

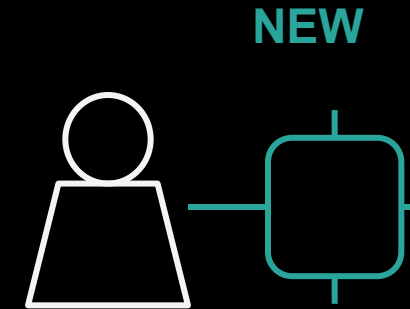
Humans add context and goals

HOW?

Models can inspect signals at scale

People know what the situation means

LLM output can be explainable & traceable



Let AI investigate

Humans add context and goals

More Questions & Challenges

HUMAN VS LLM 'EXPERTISE'



Do humans really catch what classifiers miss?

DESIGN FOR ADVERSARIES



Attackers can target the assistant too.

Can trust calibration survive manipulation?

Let's focus on a better division of labour



- AI inspects data
- people provide context and goals
- focus on ambiguity
- design interaction principles
- human heuristics for strategic input

Slides + contact

